

ADA Demographic Methods

Monica Alexander

Week 10: Multilevel regression, poststratification, and the future???

Roadmap

- ▶ Multilevel regression
- ▶ Multilevel logistic regression
- ▶ Post-stratification
- ▶ Multilevel regression and post-stratification (MRP)
- ▶ Demography needs YOU (the future)

Motivation

- ▶ Often in demography and other social sciences we are looking at settings where there is a hierarchy of nested populations.
- ▶ Many problems have a natural hierarchy e.g.
 - ▶ patients within hospitals
 - ▶ school kids within classes within schools
 - ▶ maternal deaths within countries within regions within the world
- ▶ Want to get estimates of underlying parameters of interest (e.g. probability of dying, test score, risk of disease) accounting for the hierarchy in the data
- ▶ Hierarchical / multilevel models are a natural framework for including information at different levels of the hierarchy

Motivation

- ▶ Often we are also dealing with data that is somehow **imbalanced**
 - ▶ E.g. survey data with small samples for some subgroups
 - ▶ Global estimation with some countries that have less data
 - ▶ Subnational estimation where some areas are smaller than others
- ▶ Hierarchical / multilevel models are particularly useful in these cases

Multilevel regression

Radon example

- ▶ Radon is a naturally occurring radioactive gas.
- ▶ Its decay products are also radioactive; in high concentrations, they can cause lung cancer (several 1000 deaths/year in the USA).
- ▶ Radon levels vary greatly across US homes.
- ▶ Data: radon measurements in over 80K houses throughout the US.
- ▶ Hierarchy: houses observed in counties.
- ▶ Potential predictors: floor (basement or 1st floor) in the house, soil uranium level at country level.

Radon dataset

Selected rows and columns

idnum	state	county	basement	activity
1	AZ	APACHE	N	0.3
2	AZ	APACHE		0.6
3	AZ	APACHE	N	0.5
4	AZ	APACHE	N	0.6
5	AZ	APACHE	N	0.3
6	AZ	APACHE	N	1.2

- ▶ 12,777 observations from 386 counties

What might we want to estimate/predict?

What might we want to estimate/predict?

- ▶ Expected radon level in a county
- ▶ Expected radon level in a county we did not have samples for
- ▶ Predicted radon level for a newly observed house in a particular county
- ▶ ...?

Let's (re)introduce some notation

- ▶ units $i = 1, \dots, n$, the smallest items of measurement (household)
- ▶ outcome $y = (y_1, \dots, y_n)$. The unit-level outcome being measure (log radon)
- ▶ groups $j = 1, \dots, J$ (counties)
- ▶ Indexing $j[i]$ (the county for house i)
- ▶ x_i is an indicator, whether or not measurement was taken on basement (house level)
- ▶ u_j is the uranium level in the soil (county level)

A model for log radon

$$Y_i \sim N(\mu_i, \sigma^2)$$

- ▶ Note that $\mu_i = E(Y_i)$ i.e. the expected (log) radon level for a particular house i
- ▶ How to model μ_i ?
- ▶ Let's start simple (no covariates)
- ▶ Given we know house i is in county j , how can we model μ_i ?

One option: no pooling

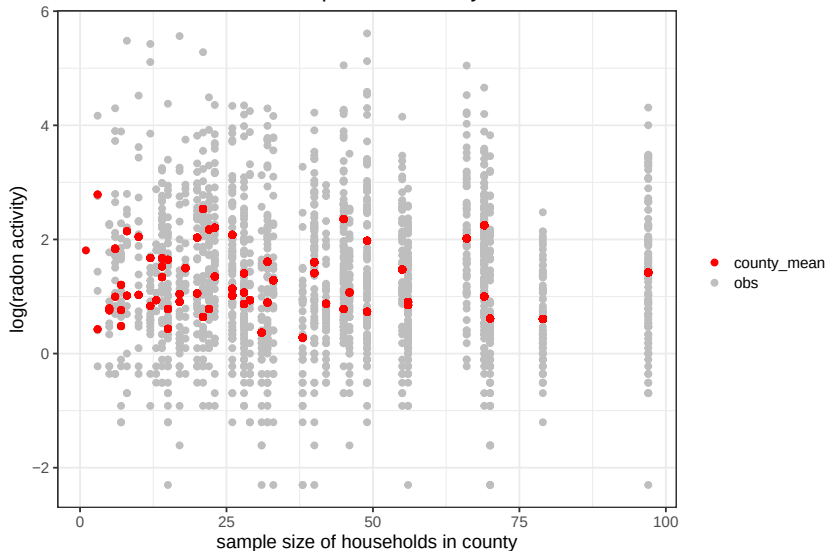
Estimate the county-level mean for each county, using only the data from that county. The model is

$$y_i \sim N\left(\alpha_{j[i]}^{\text{no pool}}, \sigma_y^2\right)$$

- ▶ The “no pool” refers to treating each county separately, i.e. no pooling of information across counties
- ▶ The most appropriate estimator for this is the county mean, i.e. $\bar{y}_j$
- ▶ I.e. the expected level of log radon for a particular house i in county j is just the mean radon level for the county

No pooling

radon measurements v sample size, Pennsylvania



What do you notice about this graph?

Another option: complete pooling

- ▶ Maybe we believe that the expected radon level for a particular house is not going to vary by county
- ▶ Use the state mean as the best estimate for the means in each county.

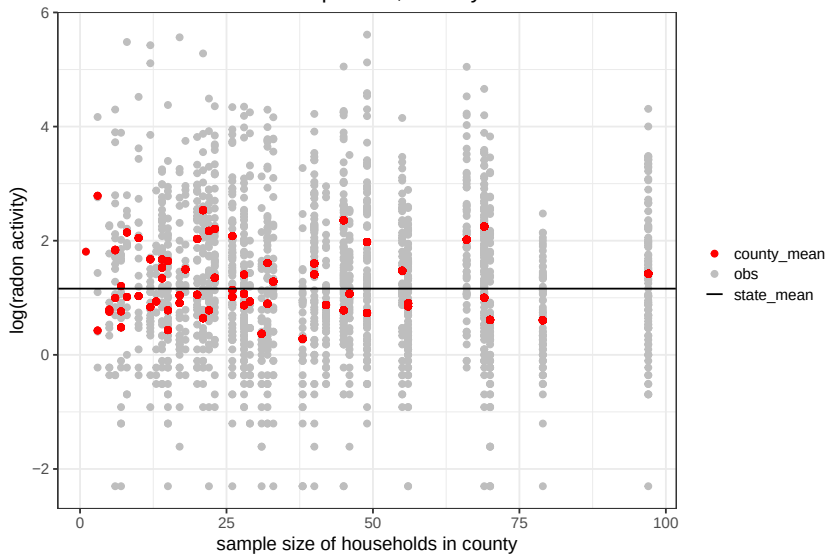
Model is

$$y_i \sim N(\mu, \sigma_y^2)$$

- ▶ i.e. expected log radon level is constant across state
- ▶ Best estimator here would just be state mean
- ▶ this is referred to “complete pooling” because information across all counties is pooled together

Complete pooling

radon measurements v sample size, Pennsylvania



Pros? Cons?

A happy medium



- ▶ Ideally we want to allow expected county radon levels to differ
- ▶ But we also want to account for information across all counties and not treat counties as separate
- ▶ A solution: partial pooling via hierarchical modeling

Another option: hierarchical model

- ▶ The expected radon level in a particular house i is
- ▶ county means α_j come from some common distribution across a state
- ▶ there are some underlying parameters governing the distribution of α 's, which are generally unknown
- ▶ middle ground between first two options, α 's are similar but not the same

Hierarchical model

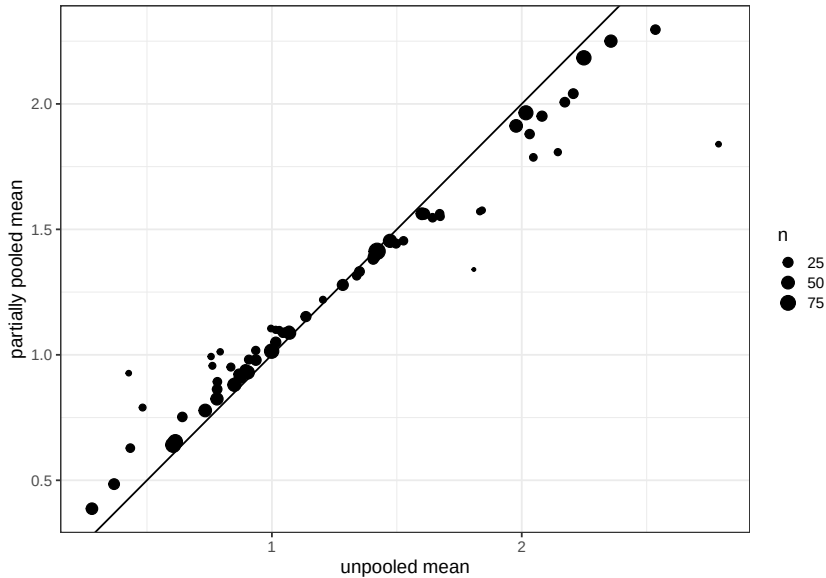
The model is

$$y_i \sim N(\alpha_{j[i]}, \sigma_y^2)$$

$$\alpha_j \sim N(\mu_\alpha, \sigma_\alpha^2)$$

- ▶ The α_j 's are themselves assumed to be from a common distribution
- ▶ μ_α and σ_α are called **hyperparameters**

The effect of partial pooling in the radon case



Where are we at

- ▶ Hierarchical models allow for 'information exchange' across groups
- ▶ Has the effect 'shrinking' group means to the overall mean
- ▶ Shrinking effect is larger when the sample size in a particular group is smaller

Adding covariates

Adding covariates

For the radon example:

- ▶ The measurements are not exactly comparable across houses because in some houses, measurements are taken in the basement, while in other houses, 1st floor measurement are taken. (This is x_i)
- ▶ Additionally, county-level uranium measurements are probably informative for across-county differences in mean levels. (This is u_j)

When adding covariates, need to think about

- ▶ what level the covariate relates to
- ▶ whether or not to model the effect hierarchically

Including covariates at the unit level

- ▶ Let x_i be the house-level first-floor indicator (with $x_i = 0$ for basements, 1 otherwise).
- ▶ This is a house-level covariate
- ▶ We can include house-level predictors in the house-level mean as follows:

$$y_i \sim N\left(\alpha_{j[i]} + \beta x_i, \sigma_y^2\right), \text{ for } i = 1, 2, \dots, n$$

$$\alpha_j \sim N\left(\mu_\alpha, \sigma_\alpha^2\right), \text{ for } j = 1, 2, \dots, J$$

Note: we have varying intercepts but a constant slope

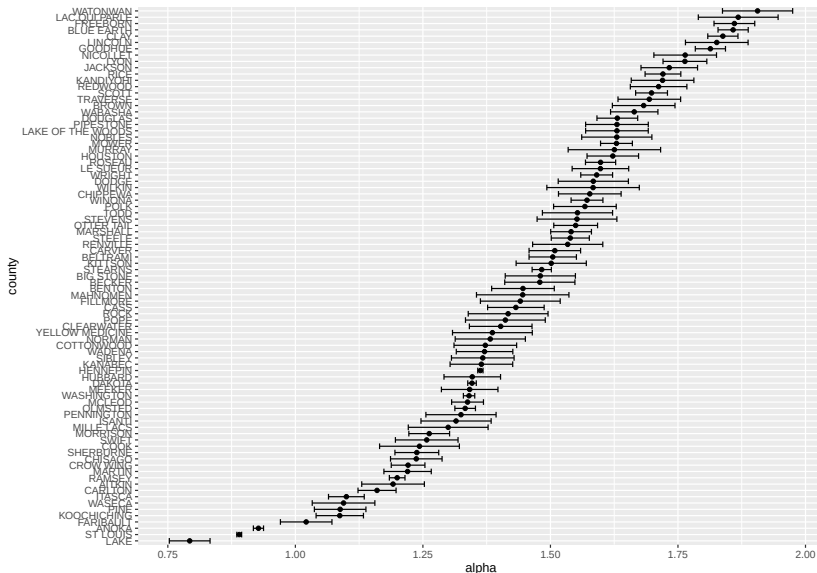
Covariates at unit level

$$y_i \sim N\left(\alpha_{j[i]} + \beta x_i, \sigma_y^2\right), \text{ for } i = 1, 2, \dots, n$$

$$\alpha_j \sim N\left(\mu_\alpha, \sigma_\alpha^2\right), \text{ for } j = 1, 2, \dots, J$$

- ▶ Estimate of β is -0.693
- ▶ Estimate of μ_α is 1.462

County-specific intercepts



Including covariates at the group level

- ▶ County-level log-uranium measurements u_j are probably informative for across-county differences in mean levels.
- ▶ We can include group-level predictors in the group-level mean as follows:

$$y_i \sim N\left(\alpha_{j[i]} + \beta x_i, \sigma_y^2\right), \text{ for } i = 1, 2, \dots, n$$

$$\alpha_j \sim N\left(\gamma_0 + \gamma_1 u_j, \sigma_\alpha^2\right), \text{ for } j = 1, 2, \dots, J$$

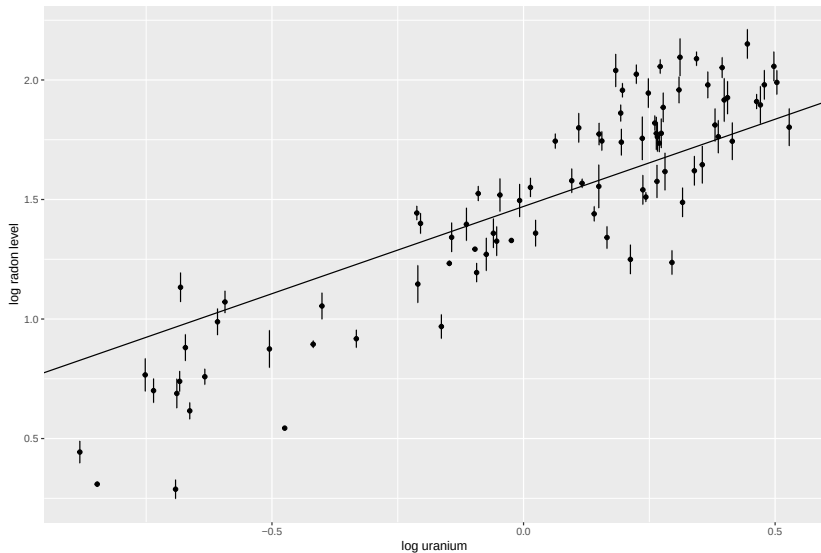
Adding covariates at group level

$$y_i \sim N\left(\alpha_{j[i]} + \beta x_i, \sigma_y^2\right), \text{ for } i = 1, 2, \dots, n$$
$$\alpha_j \sim N\left(\gamma_0 + \gamma_1 u_j, \sigma_\alpha^2\right), \text{ for } j = 1, 2, \dots, J$$

- ▶ Estimate of β is -0.668
- ▶ Estimate of γ_0 is 1.407
- ▶ Estimate of γ_1 is 0.729

County-level radon and uranium

County radon levels versus Uranium, Minnesota



Logistic regression

Logistic regression in a hierarchical context

Can easily extend these hierarchical models to model binary outcomes and the probability of an event happening by various groups of interest, e.g. in the simplest form we would have

$$y_i | \pi_i \sim \text{Bern}(\pi_i), \text{ OR}$$

$$y_i | \pi_i \sim \text{Bin}(n_i, \pi_i), \text{ if total number of trials is } n_i$$

and then

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right)$$

= some function of covariates, e.g.

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 x_i$$

and then could put some hierarchical structure on the β 's for instance.

Fertility intentions

- ▶ Using data from the 2015-2017 National Survey of Family Growth
- ▶ Gathers information on family life, marriage and divorce, pregnancy, infertility, use of contraception, and men's and women's health in the US.
- ▶ Some microdata are public:
<https://www.cdc.gov/nchs/nsfg/index.htm>

Fertility intentions

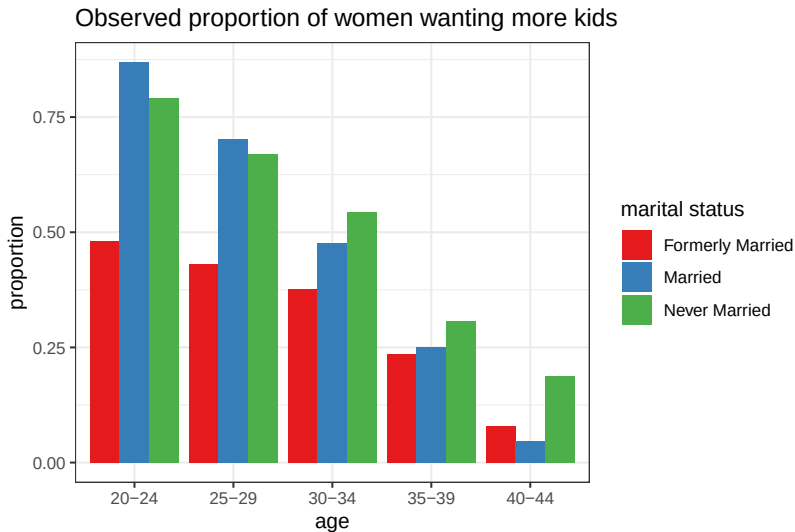
Interested in looking at fertility intentions, i.e. do women intend to have more children (and if so, how many)

- ▶ predictor of childbearing
- ▶ gaps tell you something about unmet need, due to social, economic conditions, cultural changes

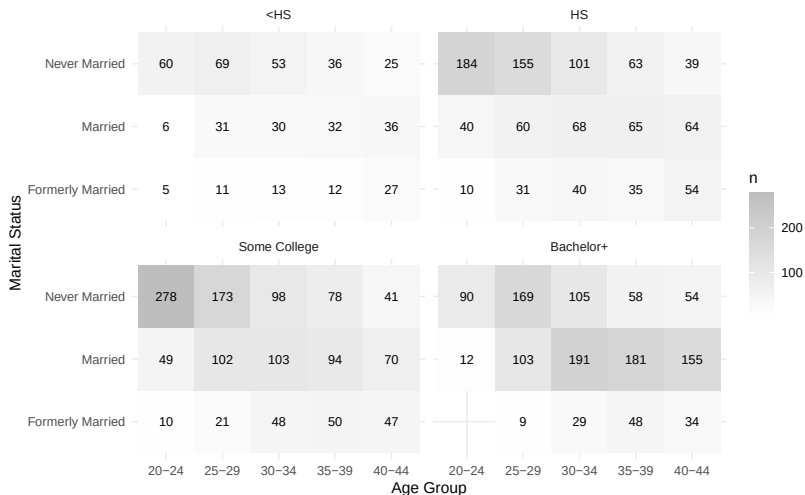
We are interested in the probability that a women wants more children.

- ▶ Individual covariates: age, education, marital status
- ▶ Possible extensions: state/region, parity (current number of children)

Some clear patterns



Sample size differences



Non-nested hierarchical model

- ▶ So far we have considered the simplest hierarchical structure of individuals i in groups j
- ▶ But individuals have different group memberships (based on age and education) that we may want to pool across.

E.g. a reasonable model set-up would be

- ▶ to have hierarchies for age and education
- ▶ an extended model that includes geography may also have county nested within state, for example

Just one hierarchical model

A hierarchical model with just age would be

$$y_i | \pi_i \sim \text{Bern}(\pi_i)$$

$$\Pr(y_i = 1) = \pi_i = \text{logit}^{-1}(\alpha_{j[i]})$$

$$\alpha_j \sim \text{N}(\mu_\alpha, \sigma_\alpha^2), \text{ for } j = 1, \dots, 5$$

Add in other effects

$$y_i | \pi_i \sim \text{Bern}(\pi_i)$$

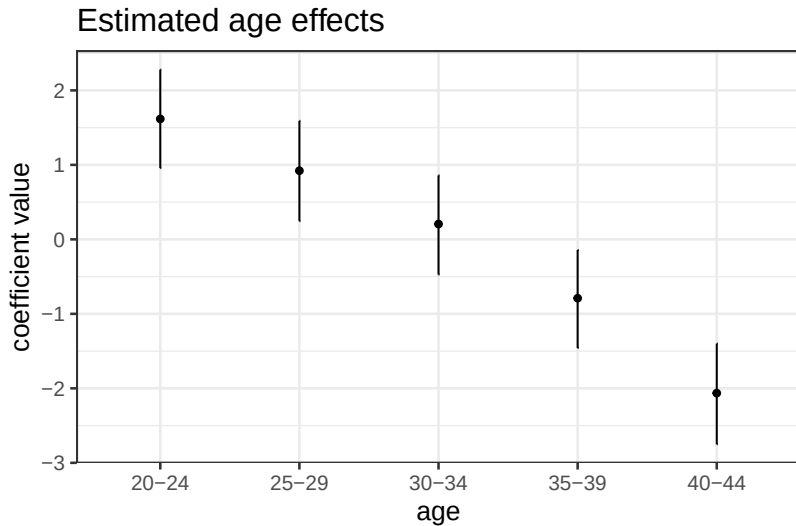
$$\pi_i = \text{logit}^{-1} \left(\beta_0 + \beta_1 \text{formerly married}_i + \beta_2 \text{married}_i + \alpha_{j[i]}^{\text{age}} + \alpha_{k[i]}^{\text{edu}} \right)$$

$$\alpha_j^{\text{age}} \sim \text{N}(0, \sigma_{\text{age}}^2), \text{ for } j = 1, \dots, 5$$

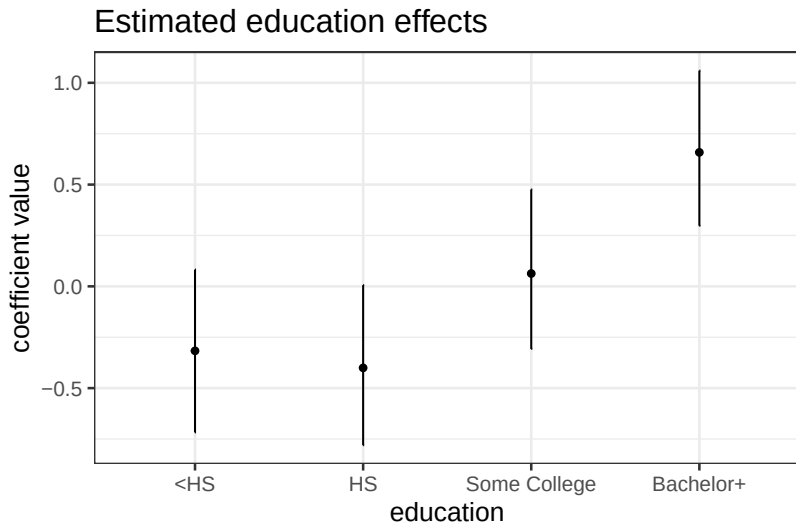
$$\alpha_k^{\text{edu}} \sim \text{N}(0, \sigma_{\text{edu}}^2), \text{ for } k = 1, \dots, 4$$

- ▶ Indexes: individual i , age j , education k
- ▶ Notice that the effects for age, educ, are centered around zero and now there is a 'global' intercept β_0
 - ▶ any non-zero means for the α s could be folded into the global intercept
- ▶ The baseline category is never married

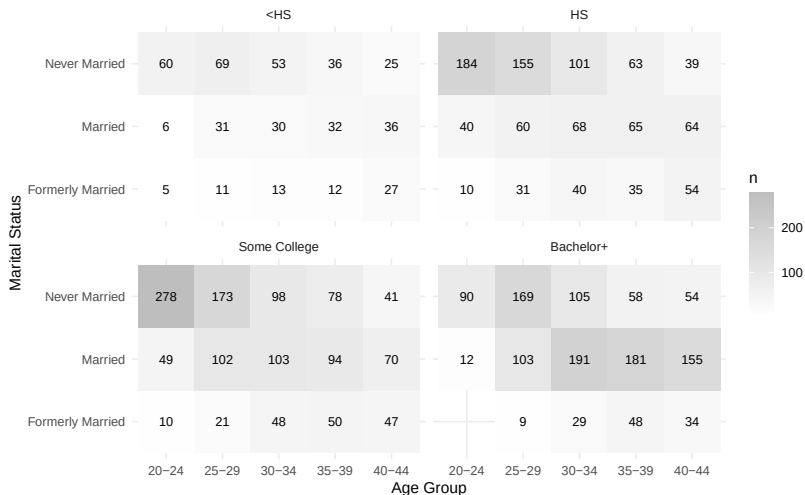
Some results



Some results

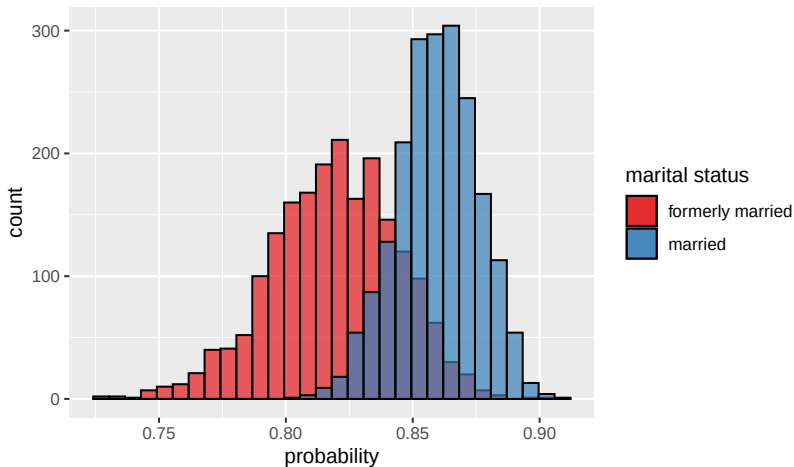


Remember we had no observations from one cell



Estimates from model

Estimated probabilities of wanting more children
20–24 year olds with Bachelor+



Summary

- ▶ Can model non-normal data in hierarchical setting
- ▶ Can have different hierarchies going on at the same time
- ▶ In the fertility intentions case, hierarchical models makes sense here because cell counts get very small: want to stabilize estimates.

Multilevel regression and poststratification

Background context

- ▶ Often want to calculate some population-level estimate of outcome of interest
- ▶ We have estimates of the outcome of interest for particular groups
- ▶ E.g. in fertility intentions example, want a national estimate of the proportion of women who would like more children, $\hat{p}_{\text{national}}$,
- ▶ Assume we can calculate estimates for each age, education, and marital status group, $\hat{p}_g$ for $g = 1, 2, \dots, G$ where $G = 3 \times 4 \times 5$ by just taking the proportions from the survey

How to generate population-level estimates?

Possibilities:

1. Simple average

$$\hat{p}_{\text{national}} = \frac{1}{G} \sum_{g=1}^G \hat{p}_g$$

2. Use survey weights (if available)

$$\hat{p}_{\text{national}} = \sum_{g=1}^G w_g \cdot \hat{p}_g$$

3. Post-stratification

- If we have another dataset that had counts of women in each group g , call them N_g , then we could calculate post-stratified estimates

$$\hat{p}_{\text{national}} = \frac{1}{\sum_g N_g} \sum_{g=1}^G N_g \cdot \hat{p}_g$$

Another option: MRP

4. Multilevel regression and post-stratification (MRP)

- ▶ Instead of using raw survey estimates of the proportion in each group ($\hat{p}_g$), we could use our modelled estimates from the multilevel (hierarchical) regression, call them $\hat{p}_g^{MR}$. The national MRP estimate is then

$$\hat{p}_{\text{national}}^{MRP} = \frac{1}{\sum_g N_g} \sum_{g=1}^G N_g \cdot \hat{p}_g^{MR}$$

- ▶ What are the advantages of the MRP estimate?

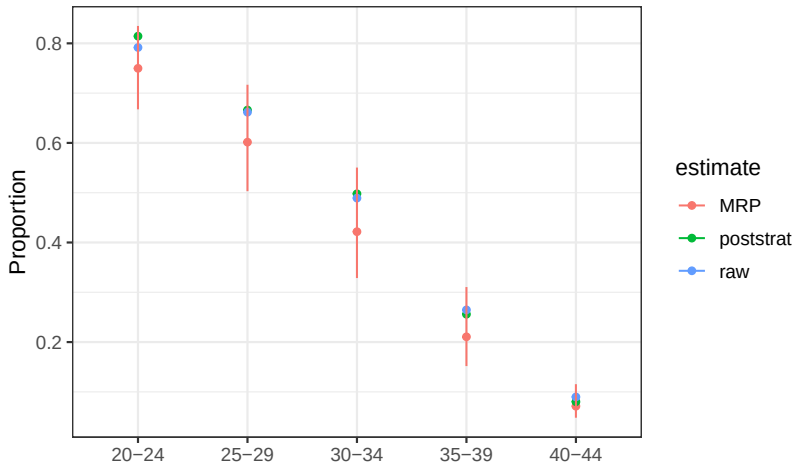
MRP

Note that we can also calculate MRP estimates across groups, for example, by age group a :

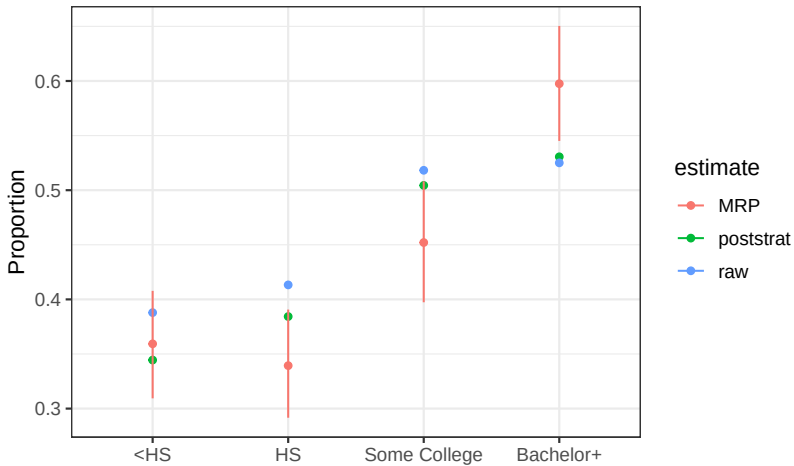
$$\hat{p}_a^{MRP} = \frac{1}{N_a} \sum_{k=1}^K N_{k,a} \cdot \hat{p}_{k,a}^{MR}$$

- ▶ N_a is number of people in age group a based on post-stratification dataset
- ▶ The k index refers to the number of education/marital status groups $k = 1, 2, \dots, K$, with $K = 3 \times 4$

Proportion of women wanting more kids by age



Proportion of women wanting more kids by education



MRP

- ▶ MRP very common in political polling (YouGov)
- ▶ Increasingly common as obtaining representative samples of populations becomes more and more difficult
- ▶ But think about model assumptions (both in terms of hierarchical structure (exchangeability) and post-strat assumptions)
- ▶ Big limitation: need post-strat info

Summary, and the future

Demography, and demographic methods

- ▶ Demography is the scientific study of population processes
- ▶ We are fundamentally interested in the core components of population change: fertility, mortality, and migration, and the social phenomena that are affected by / and affect these components
- ▶ In this course, we have focused on methods to estimate demographic indicators

Core ideas in demographic methods

- ▶ Age, period, cohorts, and understanding changes across these dimensions
 - + Lexis diagrams are your friends!
- ▶ Decomposition and standardization
- ▶ Assumptions about stable populations / synthetic cohorts
- ▶ Strong empirical patterns, dimensionality reduction
- ▶ (newer): smoothing, hierarchical pooling, informative priors

What demography can offer sociology and other social sciences

- ▶ “Yes, but have you stratified by age?” / “Yes, but have you considered age effects?”
- ▶ Individual → aggregate → individual
- ▶ It's cool to be uncool (care about data quality)
- ▶ People in positions of power will always care about populations: good to get the facts straight
- ▶ Good descriptive work is powerful

What the other social sciences can offer demography

- ▶ Strengthening the link between demographic changes and societal changes
- ▶ Descriptive findings → policy implementation
- ▶ Improved individual-level modeling

The future

1. Dealing with bad data

- ▶ We have more data now, but it's often not great
- ▶ Tools and techniques to deal with non-representative, large, biased, datasets

The future

2. Linked data sources

- ▶ We now have a much larger capacity to link different datasets
- ▶ Questions about heterogeneity, clustering, repeated measures now have the ability to be answered!

The future

3. What tf do we do with genAI

- ▶ Helping v hindering research
- ▶ Evaluation and validation is key